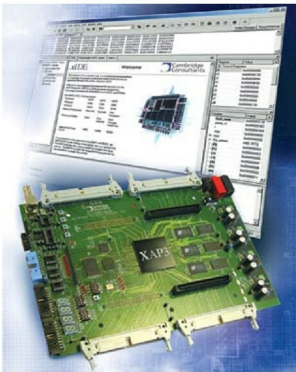


ments of hardware as well as software engineers to fix the bug when the cost is minimal.

Firmware brings your processor to a known state

"Software is critical in making the device work and achieve the desired functionality. It constitutes almost 40 per cent of any embedded device today," says Narendra Bhandari, director APAC, developer relations division, Intel Corporation. Firmware resides as the read-only software for an embedded device. "It brings the processor to a known state and provides the device driver application programming interface (API) for all the peripherals required for end application," says Gurunatham, managing director, Electro Systems Associates (ESA). Middleware stack provides communication services like USB and interfaces the application with the real-time operating system. The software for embedded devices have been increasing in complexity. The problem is exacerbated as software development may have to wait for hardware to be developed.

"You can use tools to automate the software development process by transitioning from systems engineering in SysML to performing software development in UML," explains John. There are various code composer software that help develop the software code



IDE tools for XAP3 (photo courtesy: Cambridge Consultants)

from specifications and functionalities described in hardware design. "But as the embedded product becomes complex, the automated code becomes more and more lengthy and needs a lot of customisation," expresses Gurunatham. The code needs optimisation for real-time operations.

David Almstrom, head of sales Asia Qt development frameworks, Nokia (Asia), says, "With rapid prototyping

of the software components, an iterative mode of development can be deployed to make the software more stable. This kind of development is referred as agile development."

Software development life cycle includes editing, compiling and debugging. The new trend that we see in this space is use of integrated development environment (IDE). This environment integrates many software tools, hence the definition may vary from vendor to vendor and from user to user. The basic tools offered in an IDE are compilers and debuggers. The primary

function of the compiler is to create an executable machine language code from the source code written in C, C++ or assembly code. The debugger allows you to find bugs and fix them by creating break points or watch windows. As you move towards higher and more complicated development levels, you may expect an IDE to provide tools for code analysis, high-level software design tools and time line analysis.

Wide portfolio, broad spectrum – your partner in T&M

Audio analyzers

Universal instruments flexibly configurable for measurements on analog and digital interfaces

Signal generators

Covering all instrument classes for digital, analog, baseband, RF and microwave applications

Signal and spectrum analyzers

Largest product portfolio on the market from handheld to mid-range to the fastest and most exacting analyzers available



Email : sales@rsindia.rohde-schwarz.com

Visit : www.rohde-schwarz.co.in

NEW DELHI : Tel-011-42535400, Fax: 011-42535433; BANGALORE : Tel-080-41780400, Fax No-080-41780444